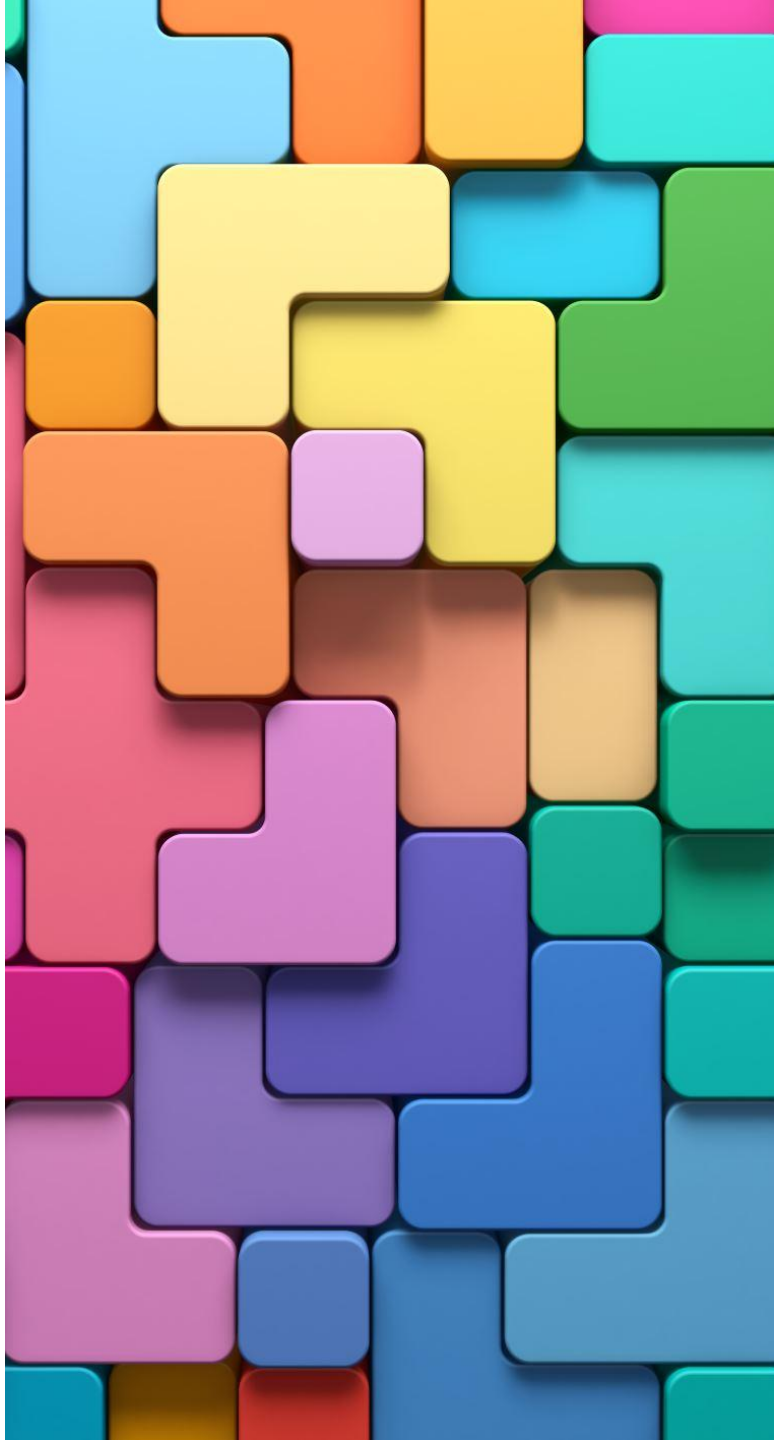
PREDICTING LOSS-MAKING FIRMS USING ADMINISTRATIVE TAX DATA



PURPOSE AND POLICY CONTEXT

Business and policy Problem:

- Kenya has persistently failed to meet Corporate Income Tax (CIT) revenue targets. The high prevalence of firms reporting losses significantly erodes the effective tax base, creating fiscal uncertainty
- Persistent loss-making firms create compliance risk, inefficient audit allocation and revenue uncertainty.

Value Proposition:

- Enables risk-based targeting in place of blanket audits by developing an automated risk scoring system that:
  - Processes firm-level Corporate Income Tax (CIT) return data following the methodology outlined in the project proposal.
  - Employs an iterative modeling approach, starting with interpretable logistic regression as a primary benchmark.
  - Leverages machine learning techniques to identify high-risk, loss-reporting firms for targeted compliance interventions.

Main Objective: To develop a supervised predictive model that estimates the probability of a firm reporting a loss, as defined in the project proposal.

Specific Objectives

- Empirically identify firm-level characteristics associated with loss reporting in CIT returns.
- Develop a supervised predictive model to estimate the probability of a firm reporting a loss.
- Assess the concentration and distribution of loss behavior across sectors and firm groups.
- Translate firm-level loss probabilities into insights on aggregate CIT revenue risk.

DATA UNDERSTANDING AND OVERVIEW

Data Source:

- The analysis is based on raw 2024 Corporate Income Tax (CIT) return data obtained from administrative tax filings. The dataset contains:
  - 313,870 firm-year observations
  - 61 variables
    - 47 numeric variables (primarily financial statement items)
    - 14 categorical variables, mainly capturing:
      - Sector classification
      - Firm size categories

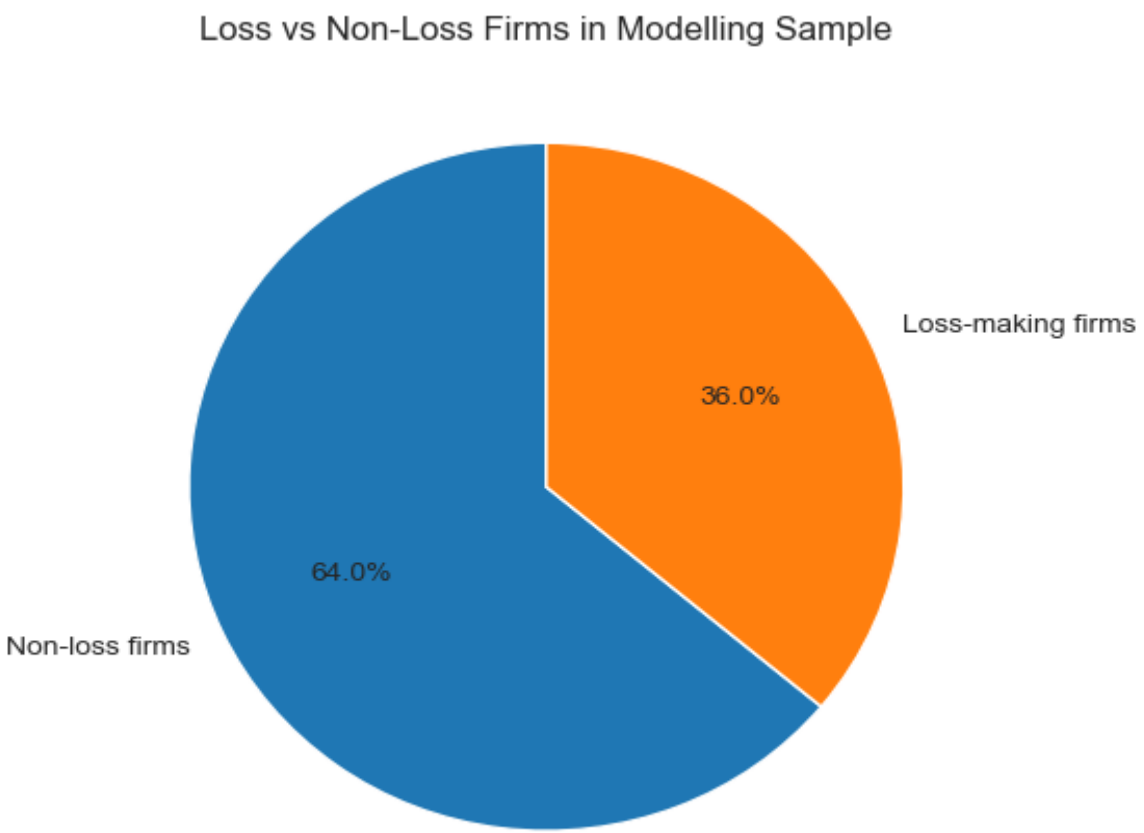
Target Variable Definition

- The target variable is defined as:
  - is_loss = 1 if Profit Before Tax (PBT) < 0
  - is_loss = 0 otherwise

Analytical scope

- Retain only firms with positive turnover, which serves as a proxy for active businesses. This excludes dormant or non-operational entities so as to reduce noise and bias the model toward structural losses rather than behavioral or economic risk.

Class Balance: moderately imbalanced a with approximately 36 loss firms and 64 non-loss firms



DATA CLEANING AND FEATURE ENGINEERING

Data Cleaning

Initial Data quality Checks:

- ❑ Quality data checks performed ; with more focus on :
- ❑ Missing values,
- ❑ Duplicates
- ❑ Coercion of mostly-numeric object colums to numeric types.
- ❑ Numeric sanity checks(Turnover, costs , profit etc)

Standardization key categorical fields

- ❑ Sector names standardized
- ❑ Rare sectors grouped for stability
- ❑ Core accounting fields aligned:TurnoverProfit before tax,Allowable deductions
- ❑ High-missing variables tagged (≥60%) and excluded from modelling

Restricting the Modelling Sample

- ❑ For comparability and validity, restriction of the modelling sample was done to firms that meet all of the following conditions:
  - ❖ Non-missing gross turnover
  - ❖ Positive gross turnover (active firms)
  - ❖ Non-missing profit before tax
- ❑ This reduces records/ data for modelling to 99491
- ❑ Class imbalance is maintained for sample.

Feature Engineering

- ❑ This stage translates raw accounting line items into economically meaningful indicators

Design Principles

- ❑ Feature construction follows four guiding principles:
  - ❖ Economic interpretability: Features reflect clear business concepts (costs, leverage)
  - ❖ Scale normalization: Ratios relative to turnover for comparability across firm sizes.
  - ❖ Leakage avoidance: No features derived from profit/target.
  - ❖ Noise robustness: Safe divisions and explicit missing-value handling.

Ratio-Based Core Features

- ❑ construct a set of the following ratios:
  - ❖ Cost-to-Turnover: Revenue absorbed by direct costs
  - ❖ Admin Cost Ratio: Overhead intensity
  - ❖ Employment Cost Ratio: Labour cost pressure
  - ❖ Financing Cost Ratio: Leverage / debt stress
  - ❖ Deductions-to-Turnover: Reliance on allowable deductions

DATA CLEANING AND FEATURE ENGINEERING

Feature Engineering...cont.

Binary Indicators for Threshold Effects

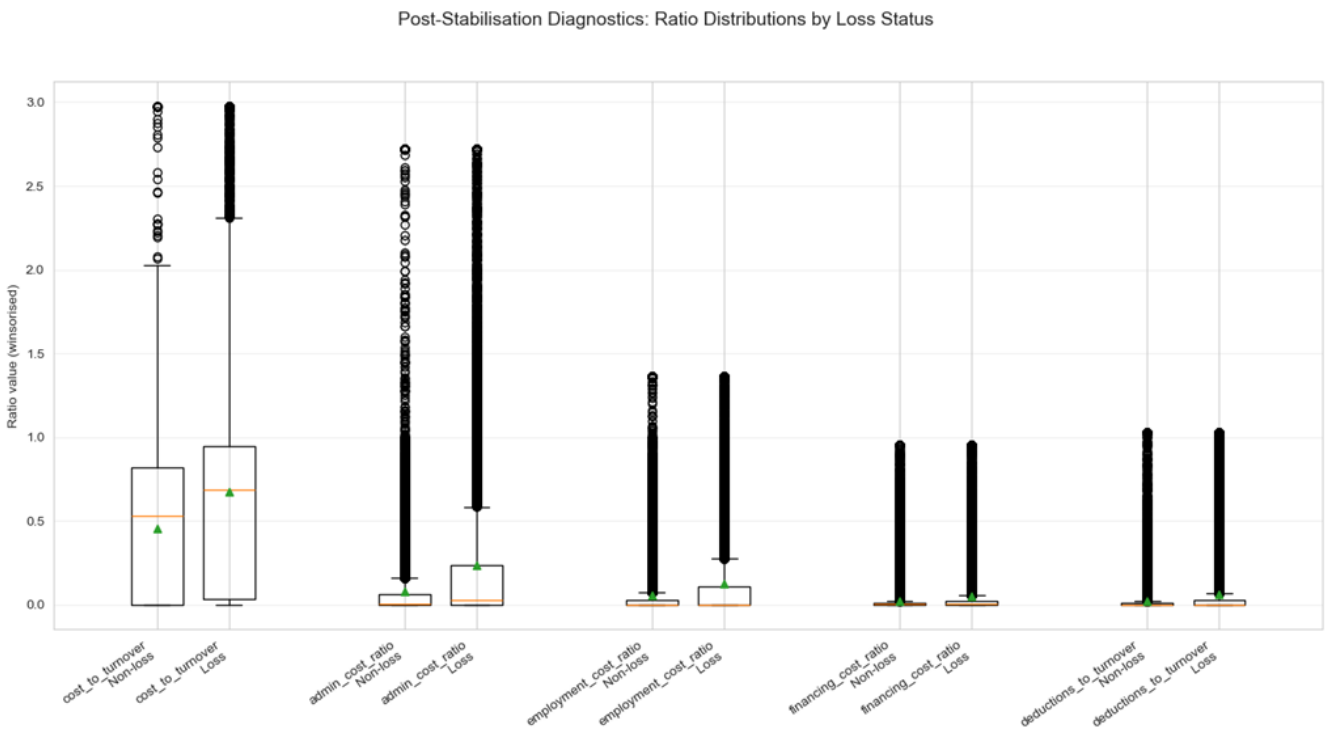
- As loss risk is often non-linear, firms near cost saturation behave differently from those with healthy margins. The following binary indicators were used:
  - High-Cost Flag: Cost-to-turnover > 90% → extreme cost pressure
  - Thin-Margin Flag: Margin 95–105% → near break-even, vulnerable to shocks

Turnover Stratification

- Firms are also grouped into turnover quartiles:Q1 (smallest firms) → Q4 (largest firms) so as to capture scale effects and identify whether loss incidence is concentrated among smaller or larger firms and to support policy-relevant segmentation

Outlier Handling

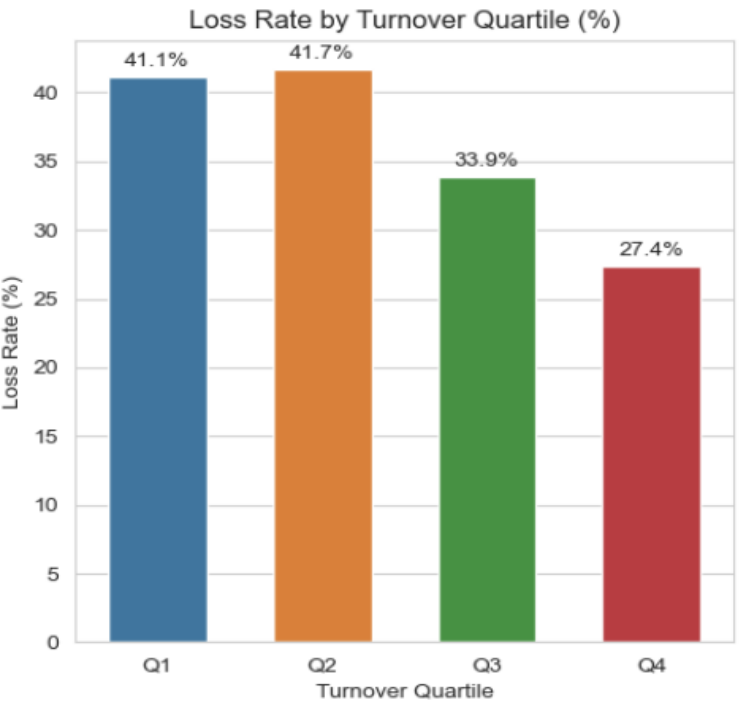
- Uncontrolled outliers can dominate model estimation, leading to unstable and misleading results.
- Each ratio feature is winsorised at the 1st and 99th Percentiles where the extreme low and high values are capped.

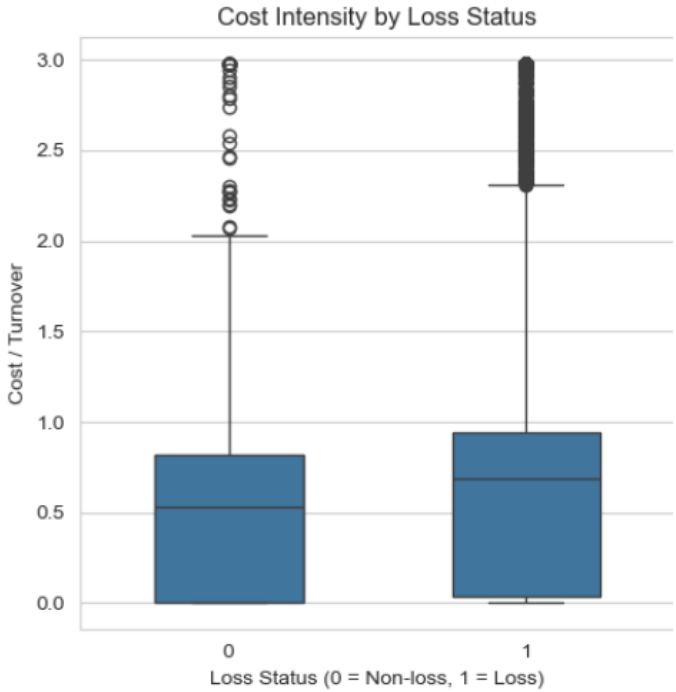


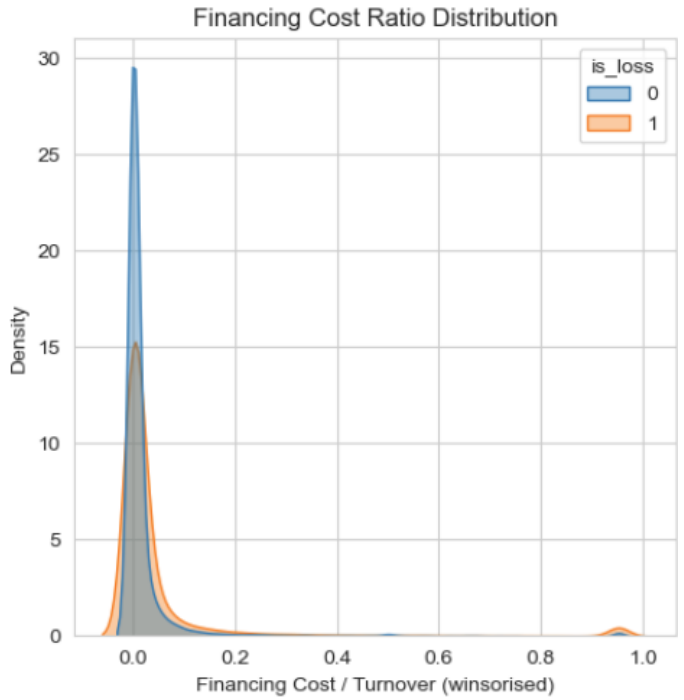
- The graph above shows that:
  - Stabilization procedure worked as intended,
  - Outliers have been effectively trimmed.
  - Distributions are reasonable and interpretable.
  - Structural differences between loss and non-loss firms are apparent.
- Additionally, data is frozen so at ensures that all subsequent modelling, reporting, and performance comparison uses exactly the same input data, eliminating accidental variation or cherry-picking.

EXPLORATORY DATA ANALYSIS – KEY PATTERNS

EDA shows strong and intuitive separation between loss and non-loss firms:







Firm size:

- ❑ Loss rates exceed 40% in the bottom turnover quartiles
- ❑ Decline steadily with size, but remain material even among large firms

Cost intensity:

- ❑ Dominant differentiator
- ❑ Loss firms show higher medians and heavy right tails indicating structural cost pressure

Financing costs:

- ❑ Right-skewed among loss firms, suggesting leverage and financing structure effects

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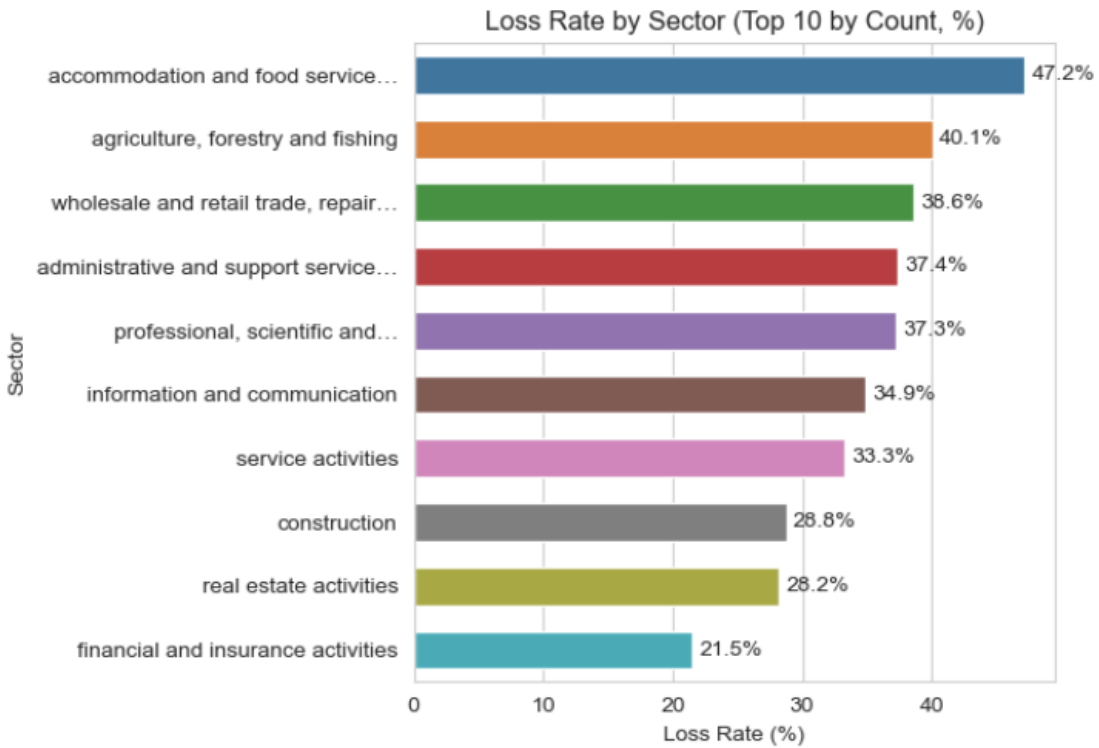
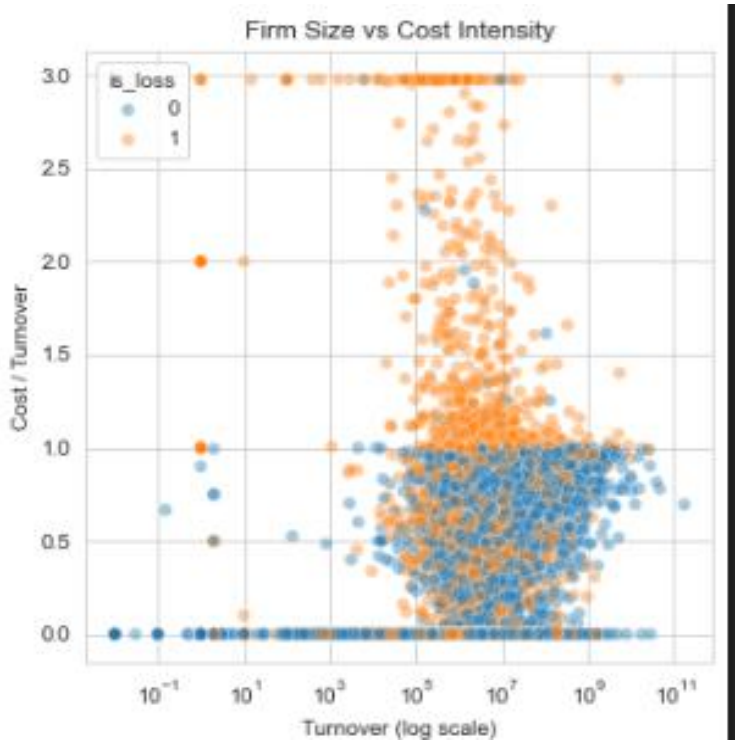




Sectoral concentration:

- ❑ Losses are concentrated in certain sectors:
  - ❖ Accommodation and food services
  - ❖ Agriculture
  - ❖ Wholesale and retail trade
  - ❖ Administrative services
- ❑ Conversely, financial and insurance sectors show lower loss rates

Firm Size vs Cost Intensity

- ❑ Confirms a non-linear relationship:
- ❑ Even large firms with elevated cost intensity are likely to report losses
- ❑ Justifies models capable of capturing non-linear effects and interactions

DATA PREPARATION AND PREPROCESSING SUMMARY

Sample and Feature summary

- Final dataset (post-cleaning and feature engineering):
- 99,332 firm observations
- Split into:
  - Training set: 74,499 observations
  - Test set: 24,833 observations
- Explanatory features: 289: Which includes engineered ratios, structural flags, and one-hot encoded categorical controls.
- Test set loss rate = 36%. This indicates moderate class imbalance, motivating metrics beyond simple accuracy such as ROC–AUC, Precision–Recall)

MODELLING STRATEGY

- Got a baseline and progressed to richer models:
  - Logistic Regression- Baseline
  - Random Forest
  - Gradient Boosting
  - XGBoost
- Identical preprocessing across all models:
- Median imputation (numeric) and Modal imputation + one-hot encoding (categorical)

Basic Model Results – Interpreting Loss Drivers

- Primary drivers of losses:
  - Cost structure
  - Financing intensity
- Sector and size are conditioning factors, not root causes
- **Policy focus:**
  - Support cost efficiency improvements
  - Ease financing constraints
  - Target small and manufacturing firms for compliance monitoring,
  - risk assessment, or targeted support

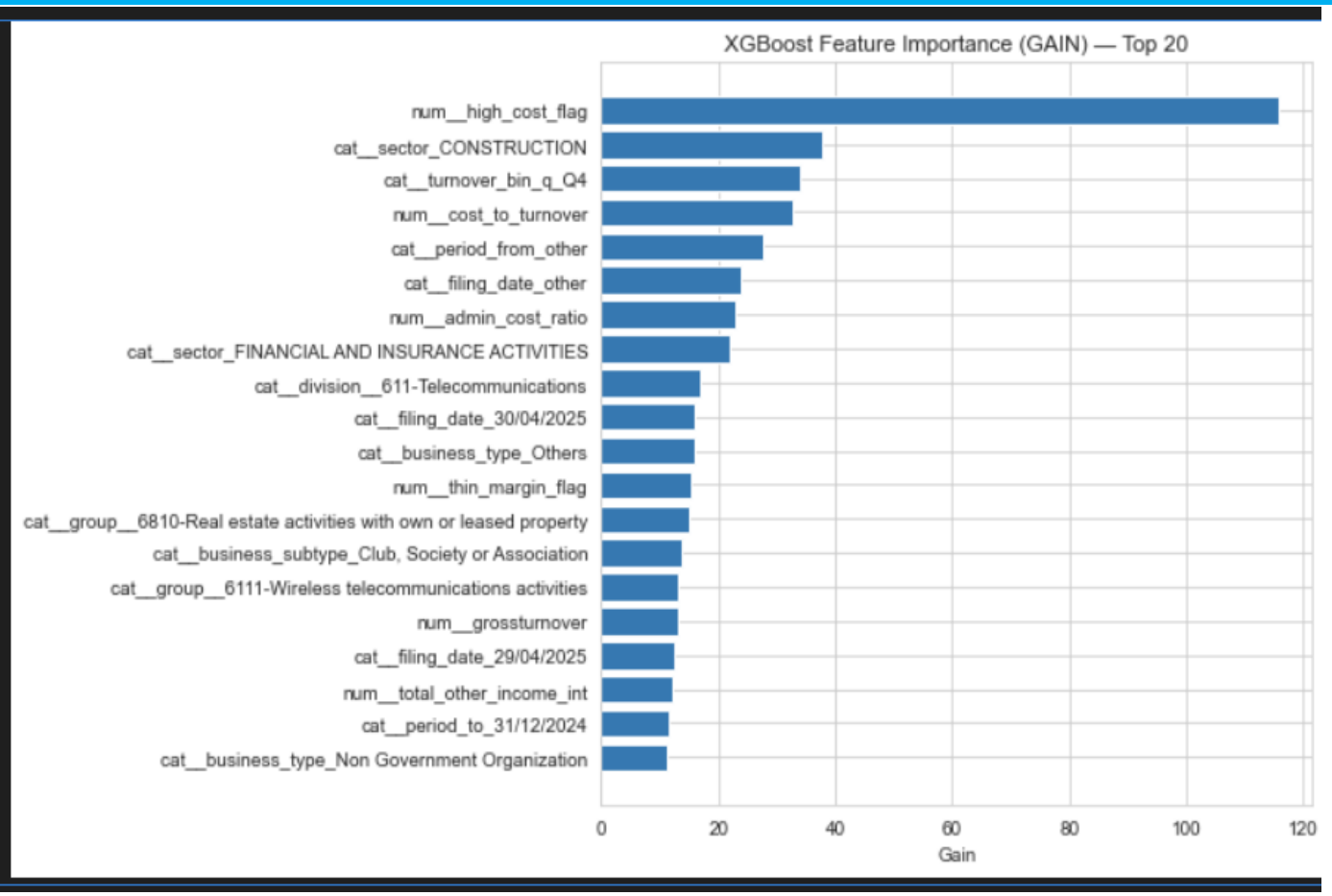
MODELS RESULTS AND COMPARISON

Model	ROC–AUC	PR–AUC	Key Takeaway
Logistic Regression (Baseline)	0.7386	0.6593	Interpretable benchmark; shows losses are driven by cost structure, size, and sector, but limited in capturing nonlinear effects.
Random Forest	0.7899	0.7252	Strong improvement over linear models; better recall of loss firms but lower precision and weaker calibration.
HistGradientBoosting	0.798	0.750	Confirms boosting is well suited; efficient and accurate, but limited control and interpretability.
XGBoost (Fast-tuned)	0.8535	0.8167	Best overall discrimination; flexible regularisation and strong precision for high-risk firms.
Optimised XGBoost (Early-Stopping)	0.8509	0.8131	Best calibrated and most stable model; preferred for operational, risk-based compliance profiling.

Key Insights:

- **XGBoost:** provides highest discrimination and precision, ideal for targeted risk profiling
- **Random Forest :** higher recall, lower precision → better for broad coverage but less targeting efficiency
- **Logistic Regression**: linear specification underperforms, limited for complex nonlinear patterns

FEATURE IMPORTANCE



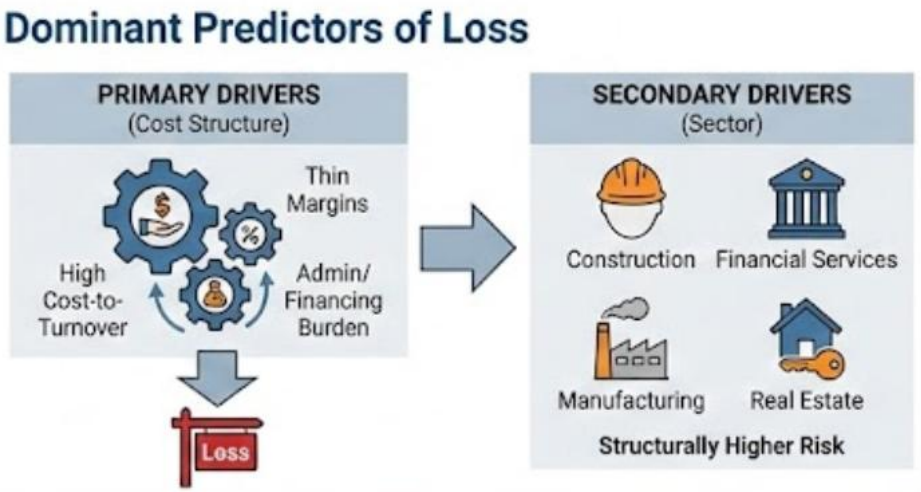
Feature Importance (GAIN):

- Measures average improvement in the objective function due to splits on each feature
- Top drivers:
  - Cost-to-turnover ratio – strongest predictor, highlights centrality of cost pressure
  - Sectoral indicators – Construction, Financial/Insurance, Manufacturing
  - Firm size quartiles – especially largest firms (Q4)
  - High-cost / thin-margin flags – viability threshold indicators
  - Financing and deduction intensity – leverage and tax adjustments
- XGBoost captures nonlinear interactions between costs, size, and sector more effectively than linear/logistic models, while maintaining economic interpretability.

STRATEGIC RECOMMENDATIONS

1. Operational Implementation

- ❑ Adopt XGBoost: Deploy the optimized model as the standard for loss-risk scoring to leverage its superior non-linear discrimination.
- ❑ Risk Banding: Shift form binary "Loss/Non-loss" flags to predicted probability bands for smarter audit prioritization and resource allocation.
- ❑ Forward-Looking Monitoring: Embed the model into revenue monitoring to identify emerging loss clusters early.



2. Policy & Compliance Re-Orientation

- ❑ Refine Risk Profiling: Move away from "firm size" as a primary risk proxy. Focus instead on cost structure indicators (eg, efficieny ratios) regardless of turnover.
- ❑ Sector-Specific Thresholds: Establish distinct risk benchmarks for high-risk sectors (Construction, Real estate, Manufacturing) where loss propensity is naturally higher.
- ❑ Target Enforcement: Complement broad policies (like minimum tax) with model-driven audits targeting firms with persistent, economically implausible loss patterns.